Diwesh Chaurasia

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Education

Rice University **Aug 2025 – Dec 2026**
Master of Computer Science, Machine Learning Specialization *Houston, TX*
Coursework: Deep Learning for Vision and Language, Statistical Machine Learning, Reinforcement Learning, Machine Learning with Graphs, Parallel Computing, Design and Analysis of Algorithms, Optimization and Complexity, Artificial Intelligence

Vellore Institute of Technology **Sep 2020 – May 2024**
Bachelor of Technology, Computer Science and Engineering *Vellore, India*

Technical Skills

Machine Learning: Transformers, CNNs, GNNs, RNNs/LSTMs, Vision-Language Models, Retrieval-Augmented Generation, Agentic Systems, Reinforcement Learning (GRPO, MDPs, Policy Optimization), LoRA, Quantization, Model Evaluation and Calibration, Statistical Testing, Decision Trees and Ensembles, Kernel Methods, PCA, K-Means and GMMs, Optimization (SGD, Adam)
Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, Hugging Face, LangGraph, vLLM, FastAPI
Programming: Python, C++, Java, SQL, JavaScript
Tools: Docker, Kubernetes, Helm, AWS, Azure, Linux, Git, GitHub Actions, OpenTelemetry, OpenMP

Experience

Rice University **Aug 2026 – Present**
Graduate Teaching Assistant, COMP 345: Foundations of Machine Learning *Houston, TX*
- Co-developed labs and problem sets for 28 lectures covering autodiff and backpropagation, transformers, self-supervised learning (SimCLR, CLIP, DINO), and reinforcement learning over MDPs.
- Led weekly recitations for an undergraduate AI cohort and advised term projects building end-to-end systems on pretrained LLMs and VLMs.

Baylor College of Medicine **Jun 2026 – Jul 2026**
Machine Learning Research Intern (Computational Biology) *Houston, TX*
- Owned as sole engineer a 12-stage pipeline fusing three heterogeneous sources into five interaction graphs, retaining 1.53M highest-precision edges under confidence thresholding and resolving entities to canonical IDs at 94.9–99.3% accuracy.
- Applied the neighborhood sampling that underpins GNN message passing to collapse a 1.53M edge corpus into seed-induced and one-hop subgraphs, encoding every node with engineered conservation features and serving interactive queries in under 15 seconds.
- Designed a consensus ranking over multi-algorithm confidence scores that surfaced the top 12% of 902 candidates as experimentally validated, and feature-based filters isolating the 29% of nodes conserved across all four target species.

Projects

signalfloor: Statistical Release Gating for LLM Evaluations *(GitHub)* **Oct 2026**
- Shipped signalfloor to PyPI, a release gate that replaces mean-score thresholds with betting confidence intervals, cutting false ships and false blocks from 48% and 42% to at most 0.3% across 90,000 held-out regression trials.
- Designed the gate to answer "needs more evidence" rather than guess, approving at most 24% of acceptable changes at 100 cases and paying for safety in sample size instead of in wrong calls.
- Built the full platform on FastAPI, PostgreSQL, OpenTelemetry, Next.js and Kubernetes, then used it on a LangGraph RAG agent to surface a prompt change that raised answer F1 by 9 points while cutting retrieval recall by 10.

Calibrating Object Hallucinations in Vision-Language Models *(GitHub)* **Jan 2026 – May 2026**
- Built a training-free calibration pipeline for Qwen2.5-VL-7B on MS COCO that verifies each entity against the image with CLIP grounding, then requests a fresh caption excluding the rejected entities, with no retraining or logit access.
- Hand-audited all 49 flagged hallucinations and found none were confirmed errors, with 29 substring artifacts, 6 color collisions and 3 annotation gaps, showing the standard CHAIR metric is unreliable at this sample size.

Chasing the Aha Moment: Reinforcement Learning for Post-Training LLMs *(GitHub)* **Oct 2026**
- Post-trained Qwen2.5-1.5B with Group Relative Policy Optimization using 4-bit NF4 quantization and rank-16 LoRA on a single T4, training in 794 seconds at 3.21 GiB peak memory with an independent replay reproducing all 64 responses exactly.
- Traced the 21/32 to 23/32 gain to answer formatting rather than better reasoning, with McNemar's test showing the difference was not significant at $p = 0.50$.
- Showed through a three-arm reward ablation that the training reward parser scored 7 of 80 correct answers as zero, pushing the policy away from correct answers instead of toward them.

Publications

Enhancing Text Summarization through Parallelization: A TF-IDF Algorithm Approach *(IEEE Xplore)* **Aug 2024**
- Profiled an extractive summarization pipeline, found tokenization and scoring dominated runtime, and parallelized those stages in C++ with OpenMP across 12 threads, cutting runtime 41% with identical output.

Honors and Achievements

Smart India Hackathon (SIH) 2023: Top-10 finalist, VIT University, building AI solutions for a Government of India challenge track.
SII Study-in-India Scholarship: Merit-based partial-tuition award for top international applicants (2020).